iGEM 2026 — Mentor Meeting Prep

MMP14 Biomarker Validation + Dry Lab Status for Potishko

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What the mentor actually asked for

Two deliverables. Do not show up without both.

1. **Biomarker validation:** Show that MMP14 is elevated in cancer blood/serum, quantify the baseline levels, and backtrack the number of detection module units needed (= answer sensitivity). Then address specificity: how selective is MMP14 for cancer vs noise?
2. **Design progress:** Confirm the dry lab has something concrete — ODE skeleton, structural scorer plan. Not "we're working on it."

MMP14 as a cancer biomarker — the numbers

Blood/serum levels

MMP14 (MT1-MMP) is a membrane-bound protease that sheds its catalytic ectodomain into circulation. It is measurable by ELISA with a detection limit of ∼0.137 ng/mL.

Cancer type	Cancer [ng/mL]	Control [ng/mL]	Fold change
Breast (invasive ductal)	16.91 ± 5.87	8.55 ± 1.66	≈2.0×
Gastric adenocarcinoma	High MMP14 ⇒	Stage III–IV correlation	Prognostic only
Colorectal	Exosomal MMP14	Trial ongoing (NCT06955767)	TBD

Key reference: Laudanski et al. (2010), *Folia Histochemica et Cytobiologica*, $p < 0.0001$.

For gastric cancer: high serum MMP14 correlates with stage III–IV ($p = 0.029$) and distant metastasis ($p = 0.022$). 5-year survival: 49.2% (low) vs 22.1% (high MMP14), HR 1.55 ($p = 0.040$).

Backcalculating units of detection module needed (sensitivity)

The mentor's question: "pull the available levels in blood to backtrack the number of units of detection module needed."

Strategy: If the minimum detectable cancer signal is at the low end of the cancer distribution (∼11 ng/mL, ≈1 SD below mean) versus the top of the healthy range (∼10 ng/mL), the **discriminating window is narrow**: Δ[MMP14] ≈ 1–6 ng/mL.

Molar conversion (MMP14 catalytic domain ≈ 18 kDa):

$$1 \text{ ng/mL} = \frac{10^{-9} \text{ g/mL}}{18000 \text{ g/mol}} = 5.56 \times 10^{-14} \text{ mol/mL} \approx 55 \text{ pM}$$

So cancer-elevated MMP14 in blood: ∼11 ng/mL ≈ **611 pM**. Healthy baseline ∼8.5 ng/mL ≈ 472 pM. The detectable difference is $\Delta \approx 140$ pM.

Backcalculation to detection units. Using the kinetics ODE from the model strategy:

$$\frac{d[\text{signal}]}{dt} = k_{\text{eff}} \cdot [S^*][G]$$

where $k_{\text{eff}} \approx 10^6$ M^{-1}s^{-1} for a 7-nt toehold (Zhang & Winfree, JACS 2009). For a 1-hour assay window, to generate detectable signal with $[E] = 140$ pM active MMP14 driving cleavage (low-substrate regime):

$$\frac{d[S]}{dt} = -\frac{k_{\text{cat}}}{K_M}[E][S] \approx -2 \times 10^5 \cdot 140 \times 10^{-12} \cdot [S]$$

This gives a pseudo-first-order rate constant $k_{\text{obs}} \approx 2.8 \times 10^{-5}$ s^{-1}. In 3600 s: ∼10% substrate cleaved. **Implication:** you need detector construct concentration at least 10–100× above background noise threshold. With a Cas12a amplification step ($k_{\text{cat,Cas}} \approx 250$ min^{-1}, LOD ∼1 fM reporter), even 1–5 pM of cleaved construct can drive signal. **Therefore: ∼1–10 nM of detection module in the assay is the ballpark target.**

Caveat to tell the mentor: MMP14 is membrane-anchored; circulating soluble MMP14 is a shed ectodomain. The SGRSENIRTA cleavage motif in our construct will be accessed by the free soluble catalytic domain, not the membrane-tethered form. This is a valid assay strategy but needs to be stated explicitly.

Specificity — is MMP14 selective enough?
Biomarker specificity (disease selectivity):

Issue	Data / mitigation
MMP14 elevated in inflammation too	Co-expression with e.g. CEA or CA15-3 increases AUC
Soluble MMP14 in healthy blood: ∼8–9 ng/mL	Not zero — assay must discriminate fold-change, not presence
Endometriosis also elevates serum MT1-MMP	Confound for female patients; context-specific claim needed
Gastric cancer correlation is prognostic, not diagnostic	Cancer type scope needs narrowing

Peptide cleavage specificity (enzyme selectivity of SGRSENIRTA):

Protease	Hydrolysis efficiency	Relevance
MMP14	79%	Target
MMP2	4%	Off-target (present in serum)
MMP9	9%	Off-target (elevated in cancer)

Source: Kridel et al. (2001), *PNAS* — cited in PMC9800621.

Honest bottom line for the mentor: MMP14 is a plausible but imperfect single biomarker. The 2× fold change in breast cancer is real but the dynamic range is narrow. The peptide cleavage is selective (79:4:9 ratio). The strongest story is **MMP14 as an activity marker** rather than a concentration marker: if the sensor detects active MMP14 (via cleavage), it's more specific than an ELISA measuring total protein. That's actually a competitive differentiator vs the 13 other cancer diagnostics teams.

Competitive landscape context

Mentor flagged 13/27 teams in the diagnostic track are doing cancer. What makes us different:

- **Most teams:** detect a biomarker concentration (ELISA-like). We detect **protease activity** — fundamentally different modality.
- **Our unique mechanism:** Protease → protein cleavage → DNA conformational unlock → strand displacement → CRISPR/fluorescent readout. This is a 4-stage signal transduction chain with a protein-DNA translator as the novel component.
- **Modularity claim:** swap in a different cleavage motif ⇒ detect a different protease. This is a platform, not a point solution. Judges love generalisable design.
- **Weakness to acknowledge proactively:** We have not yet shown that the hairpin actually locks in the constrained conformation. This is the wet lab's job. The dry lab's structural scorer (ESM2 embeddings) will give a predicted k_{open} band *before* wet lab validates. That sequencing is exactly what Best Model wants.

Dry lab deliverables — current status

What to show the mentor

Deliverable	Status	Owner
Biomarker validation (this doc)	Done (bring to meeting)	Aaran
4-stage ODE skeleton (written)	Written in Model Strategy doc	Archer / Aniu
ODE coded + runs	*Target: this week*	Archer / Aniu
Structural scorer (ESM2 plan)	Planned, not started	Aaran
ColabFold structures (ybbR/S6, A1/S6)	Done (in translator design doc)	Aaran
k_{open} estimate from scorer	Pending scorer build	Aaran
Wiki skeleton on iGEM GitLab	Not confirmed	Orlando

Honest gaps to flag

- ODE model is *designed* not *coded*. Archer and Aniu need to have at minimum a working Python script by the wet lab meeting Monday.
- The structural scorer (Aaran's workstream) has not started. ESM2 embeddings + AlphaFold3 structure needed. This is the **weakest link** for the Best Model pitch.
- k_{open} (conformational release rate) is the join between both workstreams. Until Aaran produces this, the model is illustrative not predictive.
- No-cleavage mutant construct not yet ordered. Needed as a control.

The 4-stage kinetics model (skeleton to walk mentor through)

$$\underbrace{\frac{d[S]}{dt} = -\frac{k_{\text{cat}}}{K_M}[E][S]}_{\text{Stage 1: MMP14 cleavage}} \quad \underbrace{\frac{d[I]}{dt} = \frac{k_{\text{cat}}}{K_M}[E][S] - k_{\text{open}}[I]}_{\text{Stage 2: conformational release}}$$

$$\underbrace{\frac{d[S^*]}{dt} = k_{\text{open}}[I]}_{\text{hairpin opens}} \quad \underbrace{\frac{d[P]}{dt} = k_{\text{eff}}[S^*][G]}_{\text{Stage 3: strand displacement}} \quad \underbrace{\frac{d[\text{sig}]}{dt} = \frac{k_{\text{cat,Cas}}[\text{Cas}][R]}{K_{M,\text{Cas}} + [R]}}_{\text{Stage 4: CRISPR amplification (optional)}}$$

Parameter	Value	Source
k_{cat}/K_M (MMP family)	$\approx 2 \times 10^5$ $\text{M}^{-1}\text{s}^{-1}$	Kwong et al., PNAS 2015
SGRSENIRTA selectivity	79% MMP14, 4% MMP2, 9% MMP9	Kridel et al.; PMC9800621
k_{eff} toehold exchange (7 nt)	$\approx 10^6$ $\text{M}^{-1}\text{s}^{-1}$	Zhang & Winfree, JACS 2009
k_{open}	*Unknown — Aaran's scorer*	Workstream 2 output

What the model predicts

Once k_{open} is in hand from the structural scorer, the model outputs:

- Time to half-maximum signal at physiological MMP14 concentrations (∼140–600 pM)
- Minimum enzyme concentration for detectable signal in a 1-hour assay
- Sensitivity to each parameter (Sobol sensitivity analysis) — i.e. which rate limits our detection

The Sobol analysis will almost certainly reveal that k_{open} is the dominant uncertainty. This is the honest story to tell judges.

Construct design summary (for wet lab coherence)

Primary construct: `FLAG-GGGGS-sfGFP-GGGGS-ybbR-GGGGS-SGRSENIRTA-GGGGS-S6-GGGGS-His6` (311 aa, 33.92 kDa, pI 5.95, instability index 32.9 — stable)

Mechanism:

1. ybbR and S6 anchor sites hold hairpin DNA in closed, toehold-sequestered conformation
2. MMP14 cleaves SGRSENIRTA (79% selectivity) separating ybbR-fragment from S6-fragment
3. Topological constraint released ⇒ hairpin opens ⇒ toehold exposed
4. Downstream strand-displacement gate or CRISPR readout activated

ColabFold: Both ybbR/S6 and A1/S6 constructs predicted. pLDDT high (> 80) across sfGFP domain; low (< 40) across linker/tag regions — expected. Confirms sfGFP folds correctly; linkers are disordered (good for accessibility).

Backup construct: A1/S6 (312 aa, 34.13 kDa) — more orthogonal dual-labelling for two distinct CoA-oligonucleotides.

Meeting agenda / talking points

1. **Open:** Confirm notes-taker assigned (mentor specifically asked for this). Name them at the start.
2. **Biomarker case (10 min):** Present the MMP14 serum level data. Show fold-change table. State the backcalculated sensitivity number (∼1–10 nM detector needed). Address specificity: 79% peptide selectivity, acknowledge narrow blood fold-change window, make the case that *activity detection > concentration detection.*
3. **Specificity deep-dive:** Be ready for "what about inflammation?" Answer: MMP14 is not inflammation-specific (it's in endometriosis, atherosclerosis). Counter: we're detecting *activity* not concentration; SGRSENIRTA discriminates MMP14 vs MMP2/9 at 79:4:9 ratio. Panel assay with a second marker if needed.
4. **Model status (5 min):** Walk through the 4-stage ODE skeleton. State that k_{open} is the unknown and the structural scorer (ESM2) will provide it. State the sensitivity analysis plan.
5. **Competitive differentiation (2 min):** Platform argument. Activity vs concentration. Modular motif.
6. **Gaps to flag honestly:** ODE not yet coded; structural scorer not started; need wet lab to validate hairpin locking. Ask mentor: what's his priority order?
7. **Papers to bring up:** Protein-DNA conjugation stability (mentor asked for this — see below).

Protein-DNA conjugate stability (mentor request)

Mentor asked for "stability facts of those conjugates." The ybbR/S6 pair uses Sfp phosphopantetheinyl transferase to attach CoA-oligonucleotides. Key facts:

- **Bond type:** Phosphopantetheine-serine covalent bond. Extremely stable; not hydrolysable under physiological conditions.
- **Thermal stability:** CoA-tethered oligonucleotides on proteins stable at 37°C for >24 h in standard buffers (PBS, Tris-HCl).
- **Protease stability:** The CoA linkage itself is not a protease substrate. However, the sfGFP scaffold will be degraded by MMP14 at SGRSENIRTA — by design.
- **DNA strand stability:** ssDNA is stable in non-nuclease conditions. If assaying in blood/serum, DNase activity is a concern — requires either cell-free lysate context, DNase inhibitors, or modified oligonucleotides (phosphorothioate backbone).
- **Orthogonality:** ybbR and S6 are both recognised by Sfp. For truly orthogonal dual-labelling (attaching two distinct oligos), switch to A1/S6 pair (AcpS labels A1, Sfp labels S6 — fully orthogonal).
- **Force spectroscopy data:** Literature reports CoA-protein linkage withstands >100 pN in atomic force spectroscopy — more than sufficient for the conformational constraint mechanism.

Paper to cite: The bioorthogonal protein-DNA conjugation / force spectroscopy reference cited in the Model Strategy document (needed for the structural scorer calibration too).

Questions to ask the mentor

1. Which cancer type should we anchor the biomarker claim to? (Breast cancer has the cleanest blood-level data so far.)
2. Is cell-free buffer the intended sample matrix, or are we targeting actual serum? This changes the DNase concern and the MMP14 concentration estimate.
3. What is the wet lab's timeline for expressing and purifying the construct? (Needed to plan when k_{open} from the structural scorer can be validated.)
4. Does the iGEM team have a GitLab Software Tools repository path confirmed? (Hard rule from Model Strategy doc — wiki URL must be fixed before freeze.)
5. Priority: Best Model vs Best Software Tool — are we fully committing to both or should dry lab focus on model only?